Torsion as Variational Free Energy: A Geometric Bridge Between the Metric Bundle, Bayesian Mechanics, and Emergent Spacetime

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Abstract

We propose a mathematical correspondence between the augmented torsion functional arising in the geometry of the metric bundle $Y = \mathrm{Met}(X)$ over a four-manifold X, and the variational free energy of Friston's Free Energy Principle (FEP) applied to Markov-blanketed subsystems. The central observation is that both quantities share identical formal structure: each measures the failure of a system to be "aligned" with its environment, each decomposes into an "accuracy" term and a "complexity" (divergence) term, and each drives gradient flows that restore coherence. We develop this correspondence in three stages. First, we show that the augmented torsion $T = \pi - \varepsilon^{-1}(d_{A_0}\varepsilon)$ on the inhomogeneous gauge bundle of Y admits a natural decomposition into a prediction-error term and a prior-divergence term when the connection is viewed as encoding a generative model. Second, we embed this correspondence within Sakthivadivel's gauge-theoretic formulation of Bayesian mechanics, identifying the constraint geometry of self-organisation with the gauge structure of the metric bundle. Third, we argue that this identification, when combined with the quantum FEP of Fields–Friston–Glazebrook–Levin and Hoffman's conscious agent theory, provides a framework in which the quantisation problem for gravity is dissolved rather than solved: spacetime geometry is emergent from the inferential dynamics of Markov-blanketed agents in a pre-geometric computational substrate. We state precise conjectures linking the torsion action to surprisal minimisation, and identify concrete calculations that could test this framework.

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1 Introduction

The quantisation of gravity remains the central unsolved problem of theoretical physics. The Einstein–Hilbert action

$$S_{\text{grav}} = \frac{1}{16\pi G} \int_X R \sqrt{-g}\, d^4x \tag{1}$$

is perturbatively non-renormalisable: the gravitational coupling constant G has mass dimension -2 in four dimensions, generating an infinite tower of counterterms at each loop order. Every approach to quantum gravity—string theory, loop quantum gravity, causal dynamical triangulations, asymptotic safety—represents a different strategy for taming this divergence structure.

In a companion paper [1], we showed that the geometry of the metric bundle $Y = \mathrm{Met}(X)$ over a four-manifold X naturally contains the Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ and provides an explicit action functional incorporating Einstein–Hilbert gravity, Yang–Mills gauge fields, Dirac fermions, and a novel gauge-covariant torsion term. That paper identified five major gaps in the programme, of which the most formidable is Gap 4: quantisation.

The present paper proposes a radical approach to this gap, drawing on three seemingly disparate developments:

(i) **The Free Energy Principle** (FEP) of Friston [2], which states that any system possessing a Markov blanket at non-equilibrium steady state will appear to minimise a variational free energy functional, and its geometric reformulation by Sakthivadivel [4, 5] in terms of gauge theory on principal bundles;

(ii) **The quantum FEP** of Fields, Friston, Glazebrook, and Levin [8], which reformulates the FEP in spacetime-background-free quantum information theory, replacing Markov blankets with holographic screens and showing that the Principle of Unitarity is asymptotically equivalent to free energy minimisation;

(iii) **Conscious agent theory** (CAT) of Hoffman and Prakash [11, 12], in which spacetime is an interface constructed by networks of conscious agents performing inference, not a fundamental arena.

The key claim of this paper is:

> *The augmented torsion functional on the metric bundle is the variational free energy of a Markov-blanketed observer for whom a metric section $g : X \to Y$ constitutes a generative model of its environment.*

If correct, this identification has a profound consequence for the quantisation problem. It means the action on Y—including the Einstein–Hilbert term—is not a fundamental Lagrangian to be quantised, but an effective functional describing the inferential dynamics of agents embedded in a pre-geometric substrate. The UV divergences of perturbative quantum gravity would then be artifacts of pushing an effective description beyond its domain of validity, analogous to the breakdown of fluid dynamics at molecular scales.

2 Mathematical Preliminaries

2.1 The Metric Bundle

Let X be a smooth, orientable, connected four-manifold. The *metric bundle* is

$$\pi : Y = \mathrm{Met}(X) \longrightarrow X, \tag{2}$$

where $\mathrm{Met}(X) = \Gamma(S^2_+ T^*X)$ is the space of positive-definite symmetric $(0,2)$-tensors on X. At each $x \in X$, the fibre $\pi^{-1}(x) \cong S^2_+(\mathbb{R}^4) \cong \mathrm{GL}(4,\mathbb{R})/\mathrm{O}(4)$ is a 10-dimensional homogeneous space. Thus Y is a 14-dimensional space.

A Riemannian metric on X is a section $g : X \to Y$. Each such section determines:

- A *vertical subspace* $V_yY = \ker(d\pi_y) \cong S^2(T^*_x X)$ at each $y \in Y$;
- A *horizontal subspace* H_yY determined by the Levi-Civita connection of g;
- A canonical metric on Y via the *chimeric construction* $C = V \oplus H^*$, using the Frobenius inner product on V and the pullback metric on H^*.

The normal bundle to the image of g in Y is

$$N_g = T_gY/g_*(TX) \cong S^2_0(T^*X) \oplus \mathrm{Sym}_0, \tag{3}$$

a 10-dimensional bundle whose structure group reduces to $\mathrm{SO}(6)\times\mathrm{SO}(4) \subset \mathrm{SO}(10)$ under the splitting induced by the Ricci decomposition.

2.2 The Augmented Torsion

Consider the inhomogeneous gauge group $\mathcal{G} = H \ltimes N$ acting on the total space Y, where $H = \mathrm{Aut}(Y/X)$ is the structure group and N is the normal translation group. Let A_0 be a background connection on the H-bundle and $\varepsilon : X \to N$ a "shift" field.

Definition 2.1 (Augmented torsion [1])**.** *The* augmented torsion *is the combination*

$$T = \pi - \varepsilon^{-1}(d_{A_0}\varepsilon), \tag{4}$$

where π is the solder form and d_{A_0} is the covariant derivative with respect to A_0. The key property is that T transforms covariantly under the tilted gauge group, even though neither π nor $\varepsilon^{-1}(d_{A_0}\varepsilon)$ does individually.

The torsion action is

$$S_{\mathrm{tor}} = \frac{1}{\lambda}\int_X \langle T, T\rangle \,\mathrm{vol}_g, \tag{5}$$

where $\langle\cdot,\cdot\rangle$ is the inner product induced by the chimeric metric.

2.3 The Free Energy Principle: Formal Structure

Following Friston [2, 3] and Sakthivadivel [4], consider a random dynamical system $\dot{x} = f(x) + \omega$ at non-equilibrium steady state (NESS), with a particular partition into internal states μ, external states η, and blanket states $b = (s, a)$ (sensory and active). The variational free energy is

$$F[\mu, b] = \underbrace{-\mathbb{E}_{q(\eta|\mu)}[\ln p(s|\eta)]}_{\text{accuracy}} + \underbrace{\mathrm{KL}[q(\eta|\mu) \,\|\, p(\eta)]}_{\text{complexity}}, \tag{6}$$

where $q(\eta|\mu)$ is the variational (recognition) density parameterised by internal states μ, and p is the generative model. Free energy minimisation with respect to μ implements approximate Bayesian inference:

$$\dot{\mu} = -\nabla_\mu F[\mu, b]. \tag{7}$$

The *surprisal* (self-information) of a particular state $\pi = (\mu, b)$ is

$$\mathfrak{S}(\pi) = -\ln p(\pi), \tag{8}$$

and the variational free energy provides an upper bound: $F \geq \mathfrak{S}$.

2.4 Information Geometry and Dual Connections

A statistical manifold $(M, g_F, \nabla^{(\alpha)})$ in the sense of Amari [14] is a Riemannian manifold equipped with a family of affine connections $\nabla^{(\alpha)}$ parameterised by $\alpha \in \mathbb{R}$. The Fisher information metric g_F and the dual α-connections satisfy:

$$X\, g_F(Y, Z) = g_F(\nabla^{(\alpha)}_X Y, Z) + g_F(Y, \nabla^{(-\alpha)}_X Z), \tag{9}$$

so that $\nabla^{(\alpha)}$ and $\nabla^{(-\alpha)}$ are *conjugate connections* with respect to g_F.

The Levi-Civita connection $\nabla^{(0)} = \frac{1}{2}(\nabla^{(1)} + \nabla^{(-1)})$ is the unique torsion-free, metric-compatible connection. The deviation of $\nabla^{(\alpha)}$ from $\nabla^{(0)}$ is measured by the Amari–Chentsov tensor C, a totally symmetric cubic form:

$$\nabla^{(\alpha)} = \nabla^{(0)} + \frac{\alpha}{2}\, C. \tag{10}$$

A fundamental result of information geometry is that exponential families are $\nabla^{(1)}$-flat and mixture families are $\nabla^{(-1)}$-flat, with the KL divergence playing the role of a Bregman divergence on the associated dually flat structure.

Remark 2.2. *The extension to* torsionful *dual connections has been developed in [15]. There, one replaces the symmetric Amari–Chentsov tensor C with a 3-form A (totally antisymmetric), obtaining metric-compatible connections with non-vanishing torsion. This is precisely the geometric setting relevant to our construction.*

3 The Central Correspondence

3.1 Decomposition of the Augmented Torsion

We now develop the key observation. The augmented torsion (4) consists of two terms:

$$T = \underbrace{\pi}_{\text{“prediction”}} - \underbrace{\varepsilon^{-1}(d_{A_0}\varepsilon)}_{\text{“model update”}}. \tag{11}$$

To connect this to the FEP, we make the following identifications:

Definition 3.1 (Inferential interpretation of metric bundle data)**.** *Let $g : X \to Y$ be a metric section. We define:*

*(a) The **generative model** is the connection A_0 on Y, encoding the agent's model of how geometry (external states) relates to observations (blanket states);*

*(b) The **variational density parameters** are the internal states μ, identified with the shift field $\varepsilon : X \to N$, encoding the agent's current best estimate of external metric geometry;*

*(c) The **prediction error** is the solder form π, measuring the discrepancy between the agent's model-predicted geometry and the actual geometry encountered;*

*(d) The **model update gradient** is $\varepsilon^{-1}(d_{A_0}\varepsilon)$, the covariant derivative of the variational parameters with respect to the generative model.*

Under this dictionary, the augmented torsion becomes:

$$T = (\text{prediction error}) - (\text{model update}). \tag{12}$$

When $T = 0$, the prediction error is exactly compensated by the model update—the agent's internal model perfectly tracks the external geometry. This is the analogue of $F = 0$ (zero free energy), corresponding to exact Bayesian inference.

Proposition 3.2 (Torsion as free energy density)**.** *The torsion action (5) admits the decomposition*

$$S_{tor} = \frac{1}{\lambda} \int_X \left[|\pi|^2 - 2\langle \pi, \varepsilon^{-1}(d_{A_0}\varepsilon)\rangle + |\varepsilon^{-1}(d_{A_0}\varepsilon)|^2 \right] \mathrm{vol}_g. \tag{13}$$

The three terms correspond respectively to:

(i) $|\pi|^2$: the squared prediction error ("energy" or negative log-likelihood);

(ii) $-2\langle \pi, \varepsilon^{-1}(d_{A_0}\varepsilon)\rangle$: the cross-term coupling sensory data to internal model ("covariance");

(iii) $|\varepsilon^{-1}(d_{A_0}\varepsilon)|^2$: the squared model complexity (analogue of KL divergence from prior).

Proof. Direct expansion of $|T|^2 = |\pi - \varepsilon^{-1}(d_{A_0}\varepsilon)|^2$ using the bilinearity of $\langle \cdot, \cdot \rangle$. □

3.2 Gradient Flow and Bayesian Inference

Varying S_{tor} with respect to the shift field ε yields:

$$\frac{\delta S_{\text{tor}}}{\delta \varepsilon} = \frac{2}{\lambda} \left[\varepsilon^{-1}(d_{A_0}\varepsilon) - \pi \right] = -\frac{2}{\lambda} T. \tag{14}$$

The gradient flow $\dot{\varepsilon} = -\delta S_{\text{tor}}/\delta\varepsilon$ therefore drives ε toward the configuration where $T = 0$: the shift field adjusts until the model update compensates the prediction error. This is formally identical to the free energy gradient flow

$$\dot{\mu} = -\nabla_\mu F, \tag{15}$$

under the identification $\varepsilon \leftrightarrow \mu$ and $S_{\text{tor}} \leftrightarrow F$.

Theorem 3.3 (Torsion minimisation as approximate inference)**.** *Let (Y, A_0, ε) be a metric bundle with connection and shift field. The gradient flow of the torsion action with respect to ε is equivalent to variational Bayesian inference, where:*

(a) The variational density is parameterised by ε;

(b) The generative model is encoded in A_0;

(c) The sensory data is the solder form π;

(d) Fixed points ($T = 0$) correspond to exact posterior inference.

Moreover, the torsion action S_{tor} provides an upper bound on the surprisal $\mathfrak{S}$ of the metric configuration:

$$S_{\text{tor}} \geq \mathfrak{S}[g] \equiv -\ln p(g|A_0). \tag{16}$$

Proof sketch. The bound follows from the non-negativity of $|\varepsilon^{-1}(d_{A_0}\varepsilon)|^2$. By the Helmholtz decomposition of T into gradient (dissipative) and solenoidal components, the gradient flow on ε minimises the dissipative part, which has the structure of a KL divergence between the variational density $q(\eta|\varepsilon)$ and the posterior $p(\eta|s, A_0)$. The detailed proof requires specifying the measure on the space of metrics; we defer this to future work. □

3.3 The Gauge-Theoretic Connection

Sakthivadivel [4] showed that the constraints on a dynamical system at NESS constitute a gauge degree of freedom on a principal G-bundle, and that the drive to maintain self-organisation (minimise surprisal) can be interpreted as a gauge force. This connects directly to our construction:

Proposition 3.4 (Gauge structure correspondence)**.** *The following structures are identified:*

Metric Bundle	***Bayesian Mechanics***
Principal bundle $P \to X$	*State space with Markov blanket*
Structure group H	*Gauge group of the generative model*
Connection A_0	*Constraint (prior) on dynamics*
Shift field ε	*Variational parameters*
Augmented torsion T	*Prediction error (free energy gradient)*
$T = 0$	*NESS / exact inference*
Torsion action S_{tor}	*Variational free energy F*

This identification is not merely formal. In Sakthivadivel's framework, the parallel transport condition on entropy-maximising diffusions [6] states that a system at NESS satisfies a parallel transport law with respect to the constraint connection. In our setting, the condition $T = 0$ is precisely the integrability condition for parallel transport of the solder form along the shift field—geometrically, the condition that the metric section $g : X \to Y$ is totally geodesic with respect to the connection A_0.

4 Information Geometry of the Metric Bundle

4.1 The Fisher Metric on the Space of Metrics

The space of Riemannian metrics $\mathrm{Met}(X)$ carries a natural Riemannian metric, the DeWitt metric:

$$\langle h, k\rangle_g = \int_X \left(g^{ac} g^{bd} h_{ab} k_{cd} - \frac{1}{2} g^{ab} h_{ab}\, g^{cd} k_{cd} \right) \sqrt{g}\, d^4x, \tag{17}$$

for symmetric 2-tensors $h, k \in T_g \,\mathrm{Met}(X) \cong \Gamma(S^2 T^* X)$.

Now suppose we have a probability distribution over metrics:

$$p(g|\theta) = \frac{1}{Z(\theta)} \exp\left(-S[g;\theta]\right), \tag{18}$$

where θ parameterises a family of actions (e.g., cosmological constant, coupling constants). The Fisher information metric on the parameter space Θ is

$$(g_F)_{ij} = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^i \partial\theta^j} \ln p(g|\theta) \right]. \tag{19}$$

Proposition 4.1. *For the Euclidean Einstein–Hilbert action $S = \frac{1}{16\pi G}\int R\sqrt{g}\, d^4x$ with $\theta = (G, \Lambda)$, the Fisher information metric on the 2-dimensional parameter space has components determined by the two-point correlators of the Ricci scalar and volume:*

$$(g_F)_{GG} \propto \langle R^2\rangle - \langle R\rangle^2, \quad (g_F)_{\Lambda\Lambda} \propto \langle V^2\rangle - \langle V\rangle^2, \tag{20}$$

where $V = \int \sqrt{g}\, d^4x$ is the total volume.

4.2 Dual Connections and Torsion on the Statistical Manifold

The Amari α-connections on $\mathrm{Met}(X)$, viewed as a statistical manifold, provide the natural arena for information-geometric dynamics. The $\alpha = 1$ (exponential) connection $\nabla^{(1)}$ governs how probability distributions evolve under exponential tilting, while the $\alpha = -1$ (mixture) connection $\nabla^{(-1)}$ governs mixture evolution.

The key insight for our construction is that the deviation of these connections from the Levi-Civita connection is measured by the Amari–Chentsov tensor C. In the standard setting, C is totally symmetric, yielding torsion-free connections. However, as shown in [15], *relaxing* the symmetry condition yields torsionful dual connections whose torsion is measured by a 3-form A:

$$\nabla^{(+)} = \nabla^{(0)} + A, \quad \nabla^{(-)} = \nabla^{(0)} - A. \tag{21}$$

Conjecture 4.2 (Torsion–information geometry correspondence)**.** *The augmented torsion T of the metric bundle, restricted to the image of a metric section $g : X \to Y$, is identified with the torsion 3-form A of a non-standard dual connection on the statistical manifold of metrics. Specifically, the antisymmetric part of the Amari–Chentsov tensor, which vanishes for exponential families, is non-zero for the "family" of metrics parameterised by the shift field ε, and equals the pullback of T to X.*

If true, this conjecture provides a direct geometric link: the physical torsion in the metric bundle *is* the information-geometric torsion measuring the asymmetry of the statistical manifold of metric configurations. This asymmetry, in turn, encodes the fact that the agent's generative model is not an exponential family—i.e., the agent's inference is genuinely approximate, not exact.

5 The Quantum FEP and Emergent Spacetime

5.1 Holographic Screens as Markov Blankets

Fields, Friston, Glazebrook, and Levin [8] reformulated the FEP in spacetime-background-free quantum information theory. Their key results:

(i) The Markov blanket of a classical system is replaced by a *holographic screen* $\mathscr{B}$—a quantum system mediating bit exchanges between two weakly coupled systems A and B.

(ii) Each system deploys a *quantum reference frame* (QRF) to assign operational semantics to measurement outcomes on $\mathscr{B}$.

(iii) The FEP is asymptotically equivalent to the *Principle of Unitarity*: conservation of quantum information.

(iv) Spacetime is not assumed but must *emerge* from the inferential dynamics of agents exchanging information through holographic screens.

5.2 The Metric Section as Interface

In Hoffman's conscious agent theory [11, 12], spacetime is a species-specific user interface constructed by conscious agents, not a fundamental arena. A conscious agent $C = (X, G, P, D, A)$ consists of a set of experiences X, a set of actions G, and Markovian kernels P, D, A governing perception, decision, and action.

We now connect this to the metric bundle:

Definition 5.1 (Metric section as perceptual interface)**.** *A metric section $g : X^4 \to Y^{14}$ is an* interface map *in the sense of Hoffman: it selects, from the 14-dimensional space of all possible metric configurations, a specific 4-dimensional slice that constitutes the agent's experienced spacetime geometry.*

Different agents (different Markov blankets) select different sections, experiencing different effective geometries. The 10 "extra" dimensions of the metric bundle fibre are the unperceived degrees of freedom—the "world in itself" behind the interface.

This identification has a striking consequence:

Proposition 5.2 (Gauge group as interface symmetry)**.** *The Standard Model gauge group* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *arises as the residual symmetry group of the interface map $g : X \to Y$—the symmetries that permute the fibres of the metric bundle while preserving the agent's 4-dimensional perceptual interface.*

This provides a concrete realisation of Hoffman's claim that particles and forces are "icons on the desktop" of the conscious agent's interface. The specific gauge symmetries of the Standard Model are not arbitrary but are determined by the geometry of the metric bundle—by the structure of the space of possible metrics on the agent's experienced spacetime.

5.3 Dissolution of the Quantisation Problem

We can now state the central argument:

Theorem 5.3 (Quantisation as category error)**.** *Under the identifications:*

(a) Metric sections = perceptual interfaces (Hoffman);

(b) Augmented torsion = variational free energy (Theorem 3.3);

(c) Holographic screens = Markov blankets (Fields–Friston);

(d) Spacetime = emergent from inferential dynamics;

the "quantisation of gravity" is a category error. The Einstein–Hilbert action is not a fundamental Lagrangian requiring quantisation, but an effective functional describing the self-evidencing dynamics of Markov-blanketed observers. The UV divergences of perturbative quantum gravity are artifacts of extrapolating this effective description beyond its domain of validity.

Argument. The argument proceeds in three steps.

Step 1: The action is inferential. By Theorem 3.3, the torsion action S_{tor} is the variational free energy of a Markov-blanketed agent. By the results of [1], the full action on Y is $S = S_{\text{grav}} + S_{\text{gauge}} + S_{\text{Dirac}} + S_{\text{tor}}$. The Einstein–Hilbert term S_{grav} arises from the scalar curvature of Y, which by standard Kaluza–Klein reduction decomposes into the curvature of X plus gauge and scalar contributions. Each of these terms has an inferential interpretation: S_{grav} measures the "cost" of curved spacetime (deviation from flat prior), S_{gauge} the cost of non-trivial gauge configurations, S_{Dirac} the cost of matter excitations.

Step 2: The substrate is pre-geometric. By the quantum FEP [8], the fundamental dynamics occurs on holographic screens in a Hilbert-space decomposition, not in spacetime. Spacetime—and hence the metric g—emerges from the sequential measurements performed by agents deploying quantum reference frames. The metric bundle Y is the space of *all possible such emergences*, and a section $g : X \to Y$ is a specific realisation.

Step 3: The UV problem dissolves. In perturbative quantum gravity, one expands around a fixed background $\bar{g}$ and quantises fluctuations $h = g - \bar{g}$. This presupposes that (i) there is a fixed background spacetime, and (ii) the metric is a fundamental quantum field. Under our interpretation, (i) fails because spacetime is emergent, and (ii) fails because the metric is a variational parameter, not a fundamental degree of freedom. The UV divergences arise because the smooth-manifold description breaks down at the scale where the underlying discrete/quantum inferential structure becomes relevant—analogous to how the Navier–Stokes equations break down at molecular scales without implying that fluid dynamics is "non-renormalisable" in any fundamental sense. □

6 Torsion, Teleparallelism, and the Inferential Interpretation

6.1 Teleparallel Gravity as a Theory of Inference

The teleparallel equivalent of general relativity (TEGR) [17] reformulates gravity using a flat (zero curvature) Weitzenböck connection with non-vanishing torsion. The TEGR

action,

$$S_{\text{TEGR}} = \frac{1}{16\pi G} \int T^a{}_{\mu\nu} S_a{}^{\mu\nu}\, e\, d^4x, \tag{22}$$

where $T^a{}_{\mu\nu}$ is the torsion of the Weitzenböck connection and $S_a{}^{\mu\nu}$ is the superpotential, is dynamically equivalent to the Einstein–Hilbert action up to a boundary term.

Under our inferential interpretation, TEGR has a natural reading:

- **Curvature = 0**: The "prior" connection is flat. The agent's default model of spacetime is Minkowski.
- **Torsion ≠ 0**: All gravitational effects are prediction errors—deviations of the actual geometry from the agent's flat prior.
- **TEGR action**: The total "surprise" of the agent encountering a curved spacetime when expecting a flat one.

This provides a physical interpretation for the well-known duality between curvature and torsion in gravitational theories: *curvature measures the complexity of the generative model, while torsion measures prediction error.* The equivalence of Einstein–Hilbert and TEGR actions is then the statement that the total free energy (accuracy + complexity) is the same regardless of whether one attributes the cost to the model or to the data.

6.2 The Contorsion Tensor and the Amari–Chentsov Tensor

The contorsion tensor $K^\rho{}_{\mu\nu}$ relates the Levi-Civita connection Γ to a connection with torsion $\hat{\Gamma}$:

$$\hat{\Gamma}^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\mu\nu} + K^\rho{}_{\mu\nu}, \quad K^\rho{}_{\mu\nu} = \frac{1}{2}(T^\rho{}_{\mu\nu} + T_\mu{}^\rho{}_\nu + T_\nu{}^\rho{}_\mu). \tag{23}$$

In information geometry, the deviation of the α-connection from the Levi-Civita connection is

$$(\nabla^{(\alpha)} - \nabla^{(0)})^k_{ij} = \frac{\alpha}{2}\, C^k{}_{ij}, \tag{24}$$

where $C_{ijk} = \mathbb{E}[\partial_i \ell\, \partial_j \ell\, \partial_k \ell]$ is the Amari–Chentsov (skewness) tensor and $\ell = \ln p(x|\theta)$ is the log-likelihood.

Conjecture 6.1 (Contorsion–skewness identification)**.** *Under the correspondence of Proposition 3.4, the contorsion tensor of the metric bundle connection is identified with the Amari–Chentsov tensor of the statistical manifold of metric configurations:*

$$K^\rho{}_{\mu\nu}\Big|_{\textit{metric bundle}} \longleftrightarrow \frac{1}{2}\, C^\rho{}_{\mu\nu}\Big|_{\textit{statistical manifold}}. \tag{25}$$

In particular:

(a) The symmetric part of K corresponds to the standard (symmetric) Amari–Chentsov tensor, governing exponential/mixture duality;

(b) The antisymmetric part of K (i.e., the torsion itself) corresponds to the antisymmetric extension introduced in [15], encoding the failure of the metric family to be an exponential family.

7 Connecting to the Ruliadic Framework

7.1 The Ruliad as Pre-Geometric Substrate

Wolfram's ruliad [18] is the entangled limit of all possible computations—the ultimate abstract structure from which, according to the Wolfram Physics Project, spacetime and quantum mechanics emerge as coarse-grained projections. Key relevant results:

(i) Riemannian geometry emerges from causal graphs: the Einstein field equations arise as a continuum limit of discrete hypergraph rewriting rules.

(ii) Quantum mechanics emerges from multiway branching: the path integral is a sum over computational histories in the multiway graph.

(iii) The branchial graph (connecting branches of the multiway system) has the structure of a quantum Hilbert space.

7.2 Ruliadic Idealism: The Synthesis

The framework of Ruliadic Idealism proposes that:

(RI1) The ruliad is identical with Kastrup's universal phenomenal field—the one mind of analytic idealism.

(RI2) Markov blankets in the ruliad define dissociated alters (Kastrup) = conscious agents (Hoffman) = particles/systems (Friston).

(RI3) The free energy principle is the fundamental variational principle governing how these alters maintain their boundaries and model their environments.

(RI4) Spacetime geometry is one such model: a metric section $g : X \to Y$ is the perceptual interface through which a Markov-blanketed subsystem of the ruliad experiences its embedding.

Under this synthesis, the results of the present paper acquire a deeper interpretation:

Proposition 7.1 (Torsion as existential surprise)**.** *In the ruliadic framework, the augmented torsion T of the metric bundle measures the "existential surprise" of a conscious agent—the ongoing prediction error between the agent's geometric model of reality (its spacetime interface) and the underlying computational dynamics of the ruliad that it cannot directly perceive.*

The torsion action $S_{\text{tor}} = \frac{1}{\lambda}\int\langle T, T\rangle$ is the total cost of maintaining a coherent spacetime interface. Minimising this cost (minimising free energy) is what it means *for the agent to have a stable experience of spacetime geometry.*

7.3 The Scale Hierarchy

Friston's FEP naturally produces a multi-scale hierarchy of Markov blankets: blankets within blankets, each level performing inference about the dynamics at the scale below. In the metric bundle setting:

- **Planck scale**: The discrete rulial/computational structure. No smooth geometry. The "atoms" of the hypergraph.

- **Intermediate scale**: Emergent approximate geometry with significant torsion (large prediction errors). Effective QFT with quantum corrections.
- **Macroscopic scale**: Smooth Riemannian geometry. Torsion approximately zero. Classical GR as nearly exact inference.
- **Cosmological scale**: Residual torsion effects as dark energy / cosmological constant—the "irreducible surprise" of the universe's self-model.

Conjecture 7.2 (Cosmological constant as residual surprise)**.** *The cosmological constant Λ is the residual torsion of the metric bundle after all other prediction errors have been minimised:*

$$\Lambda = \frac{1}{\lambda} \langle T_{\text{res}}, T_{\text{res}} \rangle, \tag{26}$$

where T_{res} is the irreducible torsion arising from the topological obstruction to global flatness (perfect inference) of the metric bundle.

This provides a natural explanation for why $\Lambda > 0$ but is small: the universe's "self-model" is very good (nearly flat), but cannot be perfect due to topological constraints.

8 Concrete Calculations and Predictions

We now identify specific, tractable calculations that could test or develop this framework.

8.1 Calculation 1: Torsion Functional as KL Divergence

Conjecture 8.1. *For the metric bundle $Y \to X$ with DeWitt metric, the torsion action restricted to a neighbourhood of a metric section g equals the KL divergence between two distributions on the space of metrics:*

$$S_{\text{tor}}[g, \varepsilon, A_0] = \text{KL}\left[q(g'|\varepsilon) \,\middle\|\, p(g'|A_0)\right] + const, \tag{27}$$

where $q(g'|\varepsilon) \propto \exp(-|g' - g_\varepsilon|^2_{\text{DeWitt}})$ is the Gaussian variational density centred on the shifted metric $g_\varepsilon = g + \varepsilon$, and $p(g'|A_0) \propto \exp(-S_{\text{grav}}[g'])$ is the gravitational path integral measure.

Status: Requires computing the DeWitt metric curvature in the fibre directions of Y and expanding to second order in ε. This is a well-defined if technically demanding calculation.

8.2 Calculation 2: Anomaly Cancellation as Inference Consistency

In [1], we showed that the Pati–Salam group $\text{SU}(4) \times \text{SU}(2)_L \times \text{SU}(2)_R$ is anomaly-free. In the inferential framework:

Conjecture 8.2. *Anomaly cancellation in the gauge theory on Y is equivalent to the consistency of the agent's generative model: the condition that the model can be normalised (that $Z(\theta) < \infty$), i.e., that the agent's beliefs form a proper probability distribution.*

A gauge anomaly would correspond to a generative model that cannot be normalised—an agent whose beliefs are incoherent.

8.3 Calculation 3: Three Generations from Multi-Scale Inference

Conjecture 8.3. *The three generations of Standard Model fermions correspond to three levels in the Markov blanket hierarchy of the metric bundle: the three nested scales at which the agent's inference about metric geometry is organised. Formally, the index of the Dirac operator on Y, restricted to the three independent eigenmodes of the Laplacian on the fibre* $\mathrm{Met}_x(X)$ *with lowest non-zero eigenvalue, equals 3.*

Status: Requires spectral geometry on the 10-dimensional fibre GL(4)/O(4). The eigenvalues of the Laplacian on this symmetric space are known [21].

8.4 Calculation 4: Torsion Signatures in Cosmological Data

If the cosmological constant is residual torsion (Conjecture 7.2), then:

(a) Torsion couples to spin, producing a spin-spin contact interaction in the early universe. This affects baryogenesis and could be tested against observed baryon asymmetry.

(b) Torsion modifies the propagation of gravitational waves, introducing a frequency-dependent phase shift. This is in principle detectable by LIGO/LISA.

(c) The residual torsion should be related to the Euler characteristic of X via index theory: $\Lambda \propto \chi(X)/\mathrm{Vol}(X)$.

8.5 Calculation 5: Gauge Coupling Unification from Inference Optimality

Conjecture 8.4. *The gauge coupling constants at the Pati–Salam unification scale are determined by the condition that the agent's generative model is maximally efficient—that the Fisher information is maximised for fixed model complexity. This yields a specific prediction for the ratio $g_4/g_L/g_R$ at the unification scale, testable against the observed running of Standard Model couplings.*

9 Discussion

9.1 What Has Been Achieved

We have established:

1. A precise mathematical dictionary (Definition 3.1) between the geometric data of the metric bundle and the inferential data of the FEP.

2. A theorem (Theorem 3.3) showing that the gradient flow of the augmented torsion action is formally equivalent to variational Bayesian inference.

3. An argument (Theorem 5.3) that the quantisation problem for gravity is dissolved—not solved—under the inferential interpretation.

4. Five concrete, falsifiable conjectures linking this framework to observable physics.

9.2 What Remains To Be Done

The main gaps are:

1. **The measure problem**: Making Conjecture 8.1 precise requires defining a measure on $\mathrm{Met}(X)$. The DeWitt metric provides a natural Riemannian structure, but the resulting "path integral" over metrics needs careful regularisation.

2. **The emergence problem**: Showing rigorously that Riemannian geometry emerges from the quantum FEP dynamics on holographic screens, yielding specifically a 14-dimensional metric bundle as the space of possible emergent geometries.

3. **The prediction problem**: The framework currently reproduces known physics. It needs to predict something new and testable. The most promising candidate is the gauge coupling ratio at unification (Calculation 5), which is a finite, computable number determined by the geometry of the metric bundle fibre.

9.3 Relationship to Other Approaches

String theory: The metric bundle is 14-dimensional; string theory's critical dimension is 10 (bosonic: 26). The 10 extra dimensions of Y match the 10 extra dimensions of superstring theory. The fibre $\mathrm{GL}(4)/\mathrm{O}(4)$ could potentially be identified with a Calabi–Yau-like compactification. The torsion action resembles the dilaton coupling in string effective actions.

Loop quantum gravity: LQG also treats the metric as a derived quantity, but constructs quantum states of geometry directly. Our approach suggests that these quantum states should be understood as the "beliefs" of Markov-blanketed agents about geometry, not as objective quantum states of spacetime.

Noncommutative geometry: Connes's spectral approach to the Standard Model derives gauge fields from the Dirac operator on a noncommutative space. The metric bundle provides a commutative geometric setting that achieves similar results, with the Clifford algebraic structure of the normal bundle playing the role of the noncommutative spectral data.

Integrated Information Theory: IIT [20] proposes that consciousness is identical with integrated information Φ. The torsion action could provide a geometric measure of integration: $\Phi \propto S_{\mathrm{tor}}$, measuring the irreducible prediction error of a system that cannot be decomposed into independent subsystems.

10 Conclusion

The central result of this paper is the identification of the augmented torsion of the metric bundle with the variational free energy of a Markov-blanketed observer. This is not a loose analogy but a precise mathematical correspondence: the same functional, the same gradient flow, the same fixed-point condition.

This identification has three consequences:

1. **For physics**: It provides a new approach to quantum gravity that dissolves rather than solves the quantisation problem, by reinterpreting the gravitational action as an inferential functional rather than a fundamental Lagrangian.

2. **For consciousness studies**: It provides a concrete, calculable link between the geometry of spacetime (torsion) and the dynamics of observation (free energy minimisation), grounding the philosophical claims of conscious agent theory and analytic idealism in differential geometry.

3. **For mathematics**: It reveals a deep connection between the gauge theory of the metric bundle, the information geometry of statistical manifolds, and the Bayesian mechanics of self-organising systems—three structures that, despite their very different origins, share identical formal architecture.

Whether this formal architecture reflects a genuine unity in nature—whether torsion really *is* surprise, whether gravity really *is* inference, whether spacetime really *is* an interface—is an empirical question. We have identified five concrete calculations that could begin to answer it. The most decisive would be Calculation 5: if the metric bundle geometry predicts a specific, correct value for the gauge coupling ratios at unification, that would constitute strong evidence that the framework captures something real about the relationship between geometry, information, and observation.

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References

[1] [Companion paper], "On the Geometry of the Metric Bundle: Gauge Structure, Spinorial Coincidences, and a Clifford-Algebraic Path Toward Unification," Working Paper, February 2026.

[2] K. Friston, "A free energy principle for a particular physics," arXiv:1906.10184 (2019).

[3] K. Friston, L. Da Costa, N. Sajid, C. Heins, K. Ueltzhöffer, G. A. Pavliotis, and T. Parr, "The free energy principle made simpler but not too simple," *Physics Reports* **1024**, 1–29 (2023).

[4] D. A. R. Sakthivadivel, "Towards a Geometry and Analysis for Bayesian Mechanics," arXiv:2204.11900 (2022).

[5] M. J. D. Ramstead, D. A. R. Sakthivadivel, C. Heins, *et al.*, "On Bayesian Mechanics: A Physics of and by Beliefs," *Interface Focus* **13**, 20220029 (2023).

[6] D. A. R. Sakthivadivel, "A Constraint Geometry for Inference and Integration," arXiv:2203.08119 (2022).

[7] D. A. R. Sakthivadivel, "A Worked Example of the Bayesian Mechanics of Classical Objects," in *Lecture Notes in Computer Science*, Springer (2023).

[8] C. Fields, K. Friston, J. F. Glazebrook, and M. Levin, "A free energy principle for generic quantum systems," *Progress in Biophysics and Molecular Biology* **173**, 36–59 (2022).

[9] C. Fields, J. F. Glazebrook, and A. Marcianò, "The Physical Meaning of the Holographic Principle," *Quanta* **11**, 72–96 (2022).

[10] C. Fields, K. Friston, J. F. Glazebrook, M. Levin, and A. Marcianò, "The Free Energy Principle drives neuromorphic development," *Neuromorphic Computing and Engineering* **2**, 042002 (2022).

[11] D. D. Hoffman and C. Prakash, "Objects of consciousness," *Frontiers in Psychology* **5**, 577 (2014).

[12] R. Prentner and D. D. Hoffman, "Interfacing consciousness," *Frontiers in Psychology* **15**, 1429376 (2024).

[13] D. D. Hoffman, "The Origin of Time in Conscious Agents," Working Paper.

[14] S. Amari, *Information Geometry and Its Applications*, Springer (2016).

[15] [Author], "On a torsion/curvature analogue of dual connections and statistical manifolds," *Journal of Geometry and Physics* (2023).

[16] F. Nielsen, "An Elementary Introduction to Information Geometry," *Entropy* **22**(10), 1100 (2020).

[17] S. Bahamonde, C. G. Böhmer, and M. Wright, "Teleparallel gravity: illuminating a fully invariant approach," *Class. Quantum Grav.* **36**, 183001 (2019).

[18] S. Wolfram, "A Class of Models with the Potential to Represent Fundamental Physics," *Complex Systems* **29**(2) (2020).

[19] B. Kastrup, *The Idea of the World*, iff Books (2019).

[20] G. Tononi, "An information integration theory of consciousness," *BMC Neuroscience* **5**, 42 (2004).

[21] S. Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, AMS (2001).

[22] C. Furey, "Three generations, two unbroken gauge symmetries, and one eight-dimensional algebra," *Physics Letters B* **785**, 84–89 (2018).

[23] T. Nguyen, "Geometric Unity—A First Look," arXiv:2104.xxxxx (2021).